JOURNAL OF LEARNING ANALYTICS

SOLAR SOCIETY for LEARNING ANALYTICS RESEARCH

Volume –(–), 1—14. http://dx.doi.org/xxx-xxx-xxx

Validity-Weighted Attenuation of Affective Noise in Open-Ended Teaching Feedback: An Interpretable ABSA Framework

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Abstract

Open-ended teaching evaluations are a widely used but underspecified data source for learning analytics, often carrying emotional content that is unrelated to instruction yet influences numerical ratings. Across 17,127 reviews from the anonymous, unmoderated platform RateMyProfessors.com, we find that non-pedagogical emotion ranks second only to instructional effectiveness as a driver of predicted ratings, and dominates the text of the lowest-rated reviews. We introduce Validity-Weighted Attenuation, an interpretable ABSA-based mechanism that mitigates the influence of construct-irrelevant affect on teaching-evaluation ratings under a definable pedagogical taxonomy. Each review is decomposed into clauses, classified against the taxonomy, and reduced to per-topic emotions; the modelled contribution of non-pedagogical affect is down-weighted in proportion to its measured density, with every adjustment traceable to textual evidence. Permutation controls confirm that the mechanism's behaviour depends on actual off-topic density rather than its functional form ($p < 0.001$), and expert paired comparisons **[TODO: [placeholder: multi-expert consensus results]]** agree with the relative magnitude of adjustment; fine-grained pairs exceed the permutation null ($p = [placeholder 0.007]$). Adjustments are conservative: 60% of reviews are adjusted with an average absolute rating change of 0.20 points and the overall distribution shape preserved. Adjusted ratings are not validated estimates of teaching quality. We test whether an interpretable attenuation mechanism behaves consistently with a stated validity argument, offering initial method evidence for validity-aware interpretation of teaching feedback as a complement to raw ratings in learning analytics.

Notes for Practice

- **Users:** instructors and teaching/learning support staff interpreting open feedback.
- **Use:** inspect a review's original text, topics, and adjustment trail; use it to distinguish potentially actionable pedagogical feedback from other student experience signals.
- **Do not use:** automated ranking, promotion, tenure, discipline, or any high-stakes judgment about an instructor.
- **Interpretation:** an attenuated value is a model-based alternative view under a stated relevance taxonomy; it is not a validated estimate of teaching quality.

Keywords

learning analytics; teaching feedback; validity; construct-irrelevant variance; aspect-based sentiment analysis; interpretability

Submitted: –/–/– — **Accepted:** –/–/– — **Published:** –/–/–

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1. Introduction

1.1 Open-Ended Teaching Feedback as a Learning-Analytics Trace

Open-ended teaching feedback is a widely available but analytically underspecified data source. When numerical ratings are accompanied by text, nothing specifies how the content of that text should shape interpretation of the rating. To this extent, any method that attempts to analytically inform stakeholders of the validity of such feedback must operate under cautious and auditable interpretation. However, open-ended teaching feedback presents a particular challenge for validity-oriented interpretation and current methods have been limited to primarily descriptive approaches. Sentiment analysis on Student

Evaluations of Teaching (SET) is a popular area for the development of computational and statistical techniques to gain insights into the features of student experiences, as well as feedback for institutions and faculty. SET has historically been used as an evaluative tool to inform and guide consequential administrative decisions for academic institutions (Haskell, 1997; Simpson & Siguaw, 2000; Irons et al., 2011). Despite this widespread usage, SET validity remains contested, with a long-running debate over the extent to which ratings reflect teaching effectiveness rather than extraneous influences (Langbein, 1994; Marsh & Roche, 1997). Situational factors, irrelevant content, and affective noise can act as contaminants, threatening the validity of SET as a measure of instructional experience and quality (Wongsurawat, 2011; Schiekirka & Raupach, 2015). Informal SET instruments such as RateMyProfessors.com (RMP) exhibit exacerbated contamination, due to being fully open-ended, anonymized and largely unmoderated. Thus, interpreting and acting on SET results without accounting for contamination risks misinforming stakeholders who interpret and act on these results and impacts faculty assessment (Haskell, 1997; Wongsurawat, 2011; Zumbach & Funke, 2014; Emery et al., 2003; Dowell & Neal, 1983).

Because a numerical rating collapses numerous distinct sources of student expression into a single value, it cannot by itself reveal what is driving its interpretation. The accompanying review provides a multifaceted trace through which those sources can be examined. This study focuses on that examination by treating open-ended teaching feedback as a learning-analytics trace.

1.2 The Unresolved Methodological Gap

Despite advances in computational SET analysis, existing research remains largely descriptive. Sentiments are extracted, patterns identified, but intervention is rarely attempted. As a result, contamination documented in the literature continues to distort the numerical ratings that institutions and students rely on. We introduce **validity-weighted attenuation**, an interpretable ABSA method that proportionally attenuates the modelled contribution of affect in content classified as outside of a stated pedagogical taxonomy that stakeholders have the ability to control, while retaining the original review and a review-level audit trail. This methodology enables inspection of the framework's outputs and rating-adjustment process in a consequential application where important criteria cannot be fully specified (Doshi-Velez & Kim, 2017), and where analytical outputs must connect to pedagogical interpretation and action (Gašević et al., 2015). SET represents an ideal domain for this work: ratings are consequential, contamination is well-documented, and informal platforms like RMP (being fully anonymous and unmoderated) represent a demanding stress test for validity-aware frameworks. The present work aims to establish a foundation for future validity-aware approaches in both informal and formal SET contexts, where interpretability is essential given growing concerns about reliance on black-box systems (Rybinski & Kopciuszewska, 2021; Zhuang et al., 2025). Findings from this research may also offer insights applicable to any open-ended feedback domain where numerical ratings are accompanied by unmoderated textual reviews.

Research Questions:

RQ1. How are topic-specific emotional signals associated with numerical ratings in open-ended teaching feedback?

RQ2. Can validity-weighted attenuation reduce the contribution of construct-irrelevant affect to numerical ratings while preserving relevant signals, under the taxonomy used in this study?

RQ3. Does the mechanism exhibit behaviour consistent with its validity rationale on held-out instructors, expert-labelled comparisons, and robustness controls?

To answer our research questions, we develop and apply a five-stage NLP framework to 17,127 RMP student evaluations.

1.3 Contribution and Boundaries

The primary contributions of this research are:

- To our knowledge, the first computational validity-weighted attenuation framework that directly adjusts SET ratings by proportionally down-weighting emotional content in clauses classified outside of a stated pedagogical taxonomy, where each adjustment is traceable to specific topic-level emotions and content density.
- A theoretically grounded design bridging Messick's (1995) unified validity theory with classical reliability and robust estimation principles (Cronbach, 1951; Huber, 1964), translating established psychometric justifications into a concrete computational mechanism and laying groundwork for future validity-aware frameworks.
- Initial empirical proof-of-concept validation on a large held-out-instructor sample, with expert and robustness evidence.

This framework demonstrates that construct-irrelevant variance, as defined by the stated pedagogical taxonomy, can be attenuated in an interpretable manner, empowering stakeholders with additional information for cautious interpretation of teaching feedback (Tsai & Martinez-Maldonado, 2022). Content classified as *Miscellaneous* falls outside this study's taxonomy; it is not thereby unimportant to the student experience. The resulting adjustments are not validated estimates of teaching quality and are not proposed for automated high-stakes personnel decisions such as ranking, tenure, or promotion; an audit trail alone does not provide the governance required for such use (Drachsler & Greller, 2016).

2. Background and Conceptual Framing

2.1 Validity, Construct-Irrelevant Variance, and Teaching Feedback

Messick (1995) identifies two primary threats to validity in assessment systems: construct underrepresentation and construct-irrelevant variance (CIV). Construct underrepresentation occurs when an assessment fails to adequately capture the dimensions of the construct it is intended to measure. Messick does not address student feedback systems by name; however, his unified theory explicitly extends to any assessment in which observed behaviors or responses are coded or summarized into scores, a formulation that directly encompasses SET. These validity threats are particularly pronounced when collected using informal and anonymous devices such as RMP.

Within the SET literature, this concern is well established. Marsh and Roche (1997) scrutinize the validity, bias and utility of SET in a comprehensive review. They conclude that under appropriate conditions, SET are multidimensional, reliable, and relatively valid against a variety of indicators of effective teaching. However, they condition this validity on the content and the coverage of the evaluation items, indicating that "scores averaged across an ill-defined assortment of items offer no basis for knowing what is being measured" (p. 1187). Validity is therefore treated as a property of the content an instrument elicits rather than of the instrument itself. Under this criterion, unconstrained and open-ended instruments offer no design guarantee on the composition of the content they collect.

Wongsurawat (2011) analyzes the value of open-ended SET feedback and proposes a conceptual framework for classifying comments. The author suggests that some students do not put sufficient thought and effort into evaluating their instructors. Importantly, white noise, or pedagogically irrelevant content, is identified as a source of measurement error in SET instruments. A conceptual four-quadrant framework comparing the correlation of subjective and objective comments to class averages is proposed to judge the reliability and representativeness of SET feedback. Wongsurawat indicates that comments that have little to no correlation with average evaluations should be discounted, and that administrators would benefit from metrics quantifying comment quality and reliability.

Recent psychometric work has demonstrated that CIV can be identified and mitigated in constructed-response assessments. Zhai et al. (2021) propose a Many-Facet Rasch Measurement framework that identifies three sources of CIV: the variability of assessment item scenarios, judging severity, and rater scoring sensitivity to the scenarios in tasks. Their framework mitigates these sources of CIV when estimating examinee ability. However, their approach operates under controlled conditions including expert-defined rubrics, trained raters, and a shared observable stimulus (a common video) against which responses can be evaluated. These conditions do not extend to large-scale SET systems.

Across these studies, validity threats from construct-irrelevant content, including situational confounders (Schiekirka & Raupach, 2015) and students' emotional and relational states (Li et al., 2025), are documented in both formal and informal evaluation contexts. However, existing approaches either operate under controlled conditions that do not extend to large-scale SET, or identify contamination without providing a computational mechanism to attenuate its influence on numerical ratings.

2.2 From Descriptive NLP to Validity-Oriented Interpretation

Prior research demonstrates that simple sentiment analysis on SET reviews can reliably extract coarse sentiments that have moderate-to-strong correlation with numerical ratings (Kechaou et al., 2011; Rajput et al., 2016; Iatrellis et al., 2024; Mihai, 2024). However, these early approaches combine sentiments as a single aggregate, limiting their ability to capture complex emotional context and signals, and act as inputs for validity-weighted frameworks. These limitations provide motivation for more advanced approaches that utilize deep learning and aspect-based sentiment analysis (ABSA) to analyze SET results.

Although deep learning approaches provide a finer-grained representation of emotional signals contained within SET reviews, analyzing sentiment per review aggregates multiple review aspects within a single numerical rating, in which affective noise is introduced to different extents (Rybinski & Kopciuszewska, 2021; Shuqin & Raga, 2024). Therefore, a validity-weighted attenuation framework likely requires finer-grained representation of sentiments.

Recent aspect-based approaches demonstrate that open-ended teaching feedback can be segmented and organized into distinct aspects before extracting sentiments (Bhowmik et al., 2023; Nikolic et al., 2020; Jazuli et al., 2025). However, extracting structured sentiment is not itself a validity intervention: these approaches do not specify how the pedagogical relevance of a clause should change the contribution of its affective content when interpreting the numerical rating that accompanies it. (Ren et al., 2022) extract and aggregate aspect sentiment but do not use relevance to proportionally attenuate affect's contribution to the accompanying rating; Butt et al. (2025) compare the predictive performance of BERT (Bidirectional Encoder Representations from Transformers) and LLM approaches but propose no validity-weighted, review-specific recalibration rule. Other recent work quantifies cumulative SET bias (Kopciuszewska & Rybinski, 2025) or identifies topic-specific sentiment drivers (Kim et al., 2025), but neither provides a mechanism to attenuate the influence of identified noise on numerical ratings. Research on computational estimation of comment relevance (Sorour et al., 2015; Louis & Nenkova, 2013; Xiong & Litman, 2011) demonstrates that filtering by pedagogical value improves prediction accuracy, but these detection systems have not been applied to mitigate the impact of affective noise on SET ratings.

2.3 Design Principle: Attenuation Rather Than Deletion

Addressing validity concerns within SET does not require excluding low-utility reviews. Classical reliability theory holds that measurement quality improves by reducing error variance rather than excluding observations (Cronbach, 1951). Robust estimation theory formalizes this principle, showing that down-weighting high-noise observations produces more stable and less biased estimates than exclusionary approaches (Huber, 1964). Within the SET domain, Marsh and Roche (1997) advocate for a conservative approach for bias mitigation, arguing that corrections should account for known sources of contamination while preserving students' overall evaluative intent. They further caution that corrections applied without regard to valid variance risk "throwing the validity baby out with the bias bathwater" (p. 1197), removing genuine signal alongside the contaminant. This supports a bounded, content-targeted adjustment rather than a global correction.

The attenuation mechanism itself is bounded and proportional, ensuring predictable changes to features, though the downstream model is less so. The taxonomy used in this RMP proof of concept is one generic decomposition of review content, and a different taxonomy would produce different adjustments. In formal applications, stakeholders could define or refine the categories in relation to course learning design and its documented pedagogical intent, so that the analytic lens reflects local priorities rather than a fixed computational judgment (Lockyer et al., 2013).

2.4 Research Gap

Despite computational advances in SET analysis, existing studies remain largely descriptive. To the best of our knowledge, no prior work provides a transparent computational framework that leverages topic-level sentiment to identify and attenuate construct-irrelevant emotional content within the scope of defined taxonomy and mitigate the impact of CIV on SET ratings. Since the criteria underlying such feedback systems cannot be fully specified, any framework that attempts this must treat interpretability and transparency as essential design constraints (Doshi-Velez & Kim, 2017), with outputs that are explainable and contestable (Drachsler & Greller, 2016) and connected to pedagogical interpretation and action (Gašević et al., 2015).

3. Method

3.1 Study Context and Data

As a source of data, we use SET reviews from RateMyProfessors.com (RMP), a popular professor and course review website offering anonymous use by any student. As an informal SET system, RMP represents a deliberately noisy stress test for our framework's capacity to mitigate CIV, rather than a proxy for institutional SET. RMP lists a strict "no scraping" policy. Despite attempts to contact site administrators to request permission to source data directly from the site, no permission was granted; we therefore use the open-source dataset provided by He (2020), sourced from RMP. This dataset comprises more than 19,000 individual textual SET reviews and associated numerical ratings (1–5) directed towards hundreds of professors, alongside RMP meta-tags and demographic information measured by the original researchers. For our purposes, the meta-tags and demographic information are not used.

We excluded non-English reviews, reviews containing RMP's automatic censorship string "****" (whose application was prone to error), entries with no review text or text such as "No Comment", and reviews for professors with fewer than eight total reviews. After these exclusions, the final dataset contains 17,127 reviews.

Before analysis, reviews were cleaned (encoding repair via `ftfy`, lowercase normalization, abbreviation expansion, and punctuation standardization); stemming and lemmatization were withheld to preserve contextual information for downstream models.

A schematic diagram of our framework is shown in Figure 1.

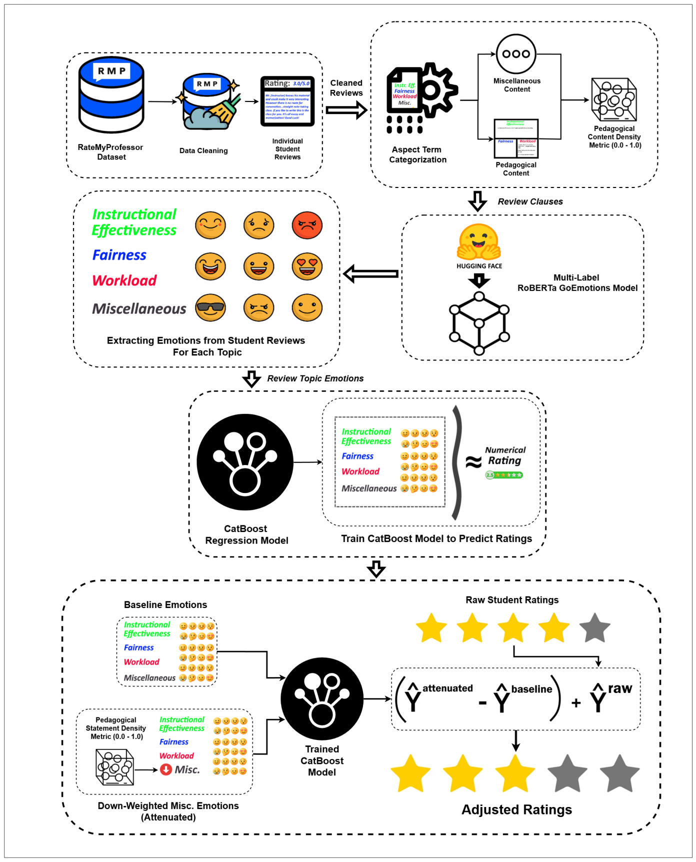

Figure 1. Schematic diagram of our framework.

3.2 Aspect-Term Categorization

Cleaned text is separated into individual clauses using `spaCy`'s `en_core_web_sm` model, which segments text based on punctuation while attempting to avoid splits on non-statement punctuation. When punctuation is sparse, two additional rules split text on commas and stop-words (but, so, yet), with aggressive stop-word separation used for reviews containing no punctuation.

The next stage in the processing pipeline assigns each review clause to one of three pedagogically relevant categories (*Instructional Effectiveness*, *Fairness & Grading*, and *Workload & Difficulty*) or to a *Miscellaneous* category. This task is formulated as an *aspect-term categorization* (ATC) problem.

Each pedagogical topic is defined using multiple short natural-language descriptions of common student expressions. These descriptions and each review clause are encoded using the pretrained sentence-embedding model `all-mpnet-base-v2`. For a clause c, cosine similarity is computed against each pedagogical topic, and the clause is assigned as follows:

$$\text{topic}(c) = \begin{cases} \arg\max\limits_{k\in\{\text{eff, fair, work}\}} s_{c,k}, & \text{if } \max_{k\in\{\text{eff, fair, work}\}} s_{c,k} > \tau, \\ \textit{Miscellaneous}, & \text{otherwise,} \end{cases} \tag{1}$$

where $s_{c,k}$ represents the cosine similarity between clause c and pedagogical topic k. We set $\tau = 0.25$, selected through validation on a development subset of manually inspected clauses to balance precision and coverage. By encoding both clause text and category descriptions as semantic vectors, the model captures meaning across the diverse phrasing and informal language common within RMP reviews.

Table 1 summarises the four-category taxonomy with representative descriptor phrases used to construct each topic embedding.

Table 1. Aspect-term categorization taxonomy and representative descriptor phrases. Clauses that do not exceed the similarity threshold $\tau = 0.25$ for any pedagogical category are assigned to *Miscellaneous*.

Category	Representative Descriptors
Instructional Effectiveness	Clear and effective explanations of concepts; lectures are well-organized and easy to follow; helps students learn and grasp the subject matter
Fairness & Grading	Grading feels fair, earned, and based on actual performance; tough but fair grader; adjusts grading to student's current ability
Workload & Difficulty	Heavy workload, lots of effort and time required; tests, exams, or quizzes are hard or demanding; easy class, light workload
Miscellaneous	Clauses below the similarity threshold τ for all pedagogical categories; includes personal remarks, social commentary, and other construct-irrelevant content

3.3 Pedagogical Density and Per-Clause Emotion Extraction

Using the categorized statements, density metrics are calculated for each review:

$$D_{\text{misc}} = \frac{\text{wordcount}_{\text{misc}}}{\text{total wordcount}}, \qquad D_k = \frac{\text{wordcount}_k}{1 + \text{wordcount}_k + \text{wordcount}_{\text{misc}}}, \qquad k \in \{\text{eff, fair, work}\}. \tag{2}$$

Here, D_{misc} represents the proportion of non-pedagogical (*Miscellaneous*) content; overall pedagogical density is therefore captured as $1 - D_{\text{misc}}$. The topic-specific density features ($D_{\text{effectiveness}}$, D_{fairness}, and D_{workload}) are also used as predictors in regression modelling, but not in the attenuation process.

Per-clause sentiment analysis is performed using `SamLowe/roberta-base-go_emotions`, a fine-tuned RoBERTa model trained on the GoEmotions dataset (Demszky et al., 2020). Each clause c is independently processed and represented by a 27-dimensional emotion vector $\mathbf{e}'_c$, where each component denotes the probability estimated by the model that the clause expresses an emotion label. The model initially produces the 28 GoEmotions labels; *Neutral* is removed because it represents the absence of strong emotional activation rather than an independent affective signal.

Using the previously obtained topic labels, clause-level emotion vectors within each review are aggregated at the topic level by computing the mean emotion vector over all clauses assigned to topic k:

$$\mathbf{E}_k = \frac{1}{|C_k|} \sum_{c \in C_k} \mathbf{e}'_c, \tag{3}$$

where C_k denotes the set of clauses in the review labelled with topic k. If no clauses are assigned to a topic, the corresponding $\mathbf{E}_k$ is imputed as a zero vector of length 27, preserving the fixed input dimensionality while indicating the absence of observed emotion for that topic.

3.4 Predicting Ratings from Topic Specific Emotions

Ratings are modelled as a function of emotions directed towards each topic and density metrics. For a single review, the input feature vector is represented as:

$$\mathbf{x}_{\text{review}} = \begin{bmatrix} \mathbf{E}_{\text{eff}}, \ \mathbf{E}_{\text{fair}}, \ \mathbf{E}_{\text{work}}, \ \mathbf{E}_{\text{misc}}, \\ D_{\text{eff}}, \ D_{\text{fair}}, \ D_{\text{work}}, \ D_{\text{misc}} \end{bmatrix} \in \mathbb{R}^{4\times 27+4} = \mathbb{R}^{112}. \tag{4}$$

Here, $\mathbf{E}_k \in \mathbb{R}^{27}$ is the aggregated emotion vector for topic k, and D_k is the corresponding density metric. For the complete dataset of $N = 17{,}127$ reviews, the feature matrix is $\mathbf{X} \in \mathbb{R}^{N\times 112}$.

To prevent data leakage and assess generalization to unseen professors, we employ an 80–20 professor-level grouped split. The CatBoost regressor is trained on 823 professors and evaluated on 206 held-out professors; all attenuation validation metrics are computed exclusively on the held-out professors. This design produces more conservative performance estimates than random splitting while reflecting the challenge of generalizing across professors in larger-scale SET scenarios.

We employ CatBoost regression, an open-source gradient-boosted tree model that captures non-linear interactions between emotional signals and topics in the sparse review representation (Hancock & Khoshgoftaar, 2020; Prokhorenkova et al., 2018; Dorogush et al., 2018). The final configuration was selected through grouped 5-fold cross-validation (GroupKFold, grouped by professor ID) on the training professors: 1000 iterations, learning rate 0.02, depth 7, and L2 leaf regularization 6; all other hyperparameters were kept at their default values. MAE was used as both the loss function and evaluation metric because it measures average absolute deviation from ratings on the 1–5 scale and is resilient to occasional outliers.

3.5 Validity-Weighted Attenuation & Rating Adjustment

Raw ratings are adjusted to mitigate the impact of CIV while preserving pedagogical signals. Emotional features present in E_{misc} are down-weighted in proportion to D_{misc}, the density of *Miscellaneous* content in the overall review. The down-weighting factor ζ is calculated as:

$$\zeta = 1 - D_{\text{misc}}. \tag{5}$$

A review with no detected *Miscellaneous* content is unchanged ($\zeta = 1$), whereas a review composed entirely of such content receives zero weight for its *Miscellaneous* emotion features ($\zeta = 0$). Intermediate densities produce proportionate attenuation. The factor ζ implements the reliability and robust-estimation rationale for down-weighting high-noise observations rather than discarding them (Cronbach, 1951; Huber, 1964).

$$\hat{E}_{\text{misc}}^{\text{weighted}} = E_{\text{misc}} \cdot \zeta. \tag{6}$$

Applying the trained CatBoost model f_{CB}, held fixed across conditions, to the baseline and attenuated feature vectors produces:

$$\begin{aligned} \hat{y}_i^{\text{attenuated}} &= f_{\text{CB}}\left(\hat{\mathbf{x}}_i^{\text{weighted}}\right), \\ \hat{y}_i^{\text{baseline}} &= f_{\text{CB}}(\mathbf{x}_i), \qquad i = 1, \ldots, N. \end{aligned} \tag{7}$$

Adjusted ratings are defined as:

$$\hat{y}_i^{\text{adjusted}} = y_i^{\text{raw}} + \left(\hat{y}_i^{\text{attenuated}} - \hat{y}_i^{\text{baseline}}\right), \quad i = 1, \ldots, N. \tag{8}$$

The prediction difference $(\hat{y}_i^{\text{attenuated}} - \hat{y}_i^{\text{baseline}})$ functions as a correction Δ, quantifying the rating shift attributable to attenuating E_{misc}. Adding this correction to the raw rating changes only the model-estimated influence of *Miscellaneous* sentiment, rather than excluding it entirely, while attempting to preserve the student's original evaluative intent.

3.5.1 Audit Trail and Worked Example

Table 2 presents a complete end-to-end trace of an individual review through the framework pipeline, covering clause extraction and categorization, per-clause emotion extraction, topic-level feature vector aggregation, validity-weighted attenuation, and final rating adjustment. Table 2 demonstrates a review with high *Miscellaneous* density (D_{misc}=0.670), where non-pedagogical sentiment dominates, receives a positive adjustment of $+0.427$.

When a review's passage through the framework is presented stage by stage, stakeholders can inspect and challenge individual outputs and overall adjustments.

4. Results

4.1 Corpus and Model Behaviour

4.1.1 ATC Topic Structure

Table 3 summarises topic frequencies after ATC. *Instructional Effectiveness* is the most frequent topic at both clause and review level, followed by *Miscellaneous* and *Workload*, with *Fairness* trailing substantially. The prominence of *Miscellaneous* clauses indicates that a substantial portion of student discourse falls outside pedagogical dimensions.

Table 3. Clause and review-level topic frequencies.

Topic	Clause	Review	Avg./Review
Instructional Eff.	25,845	13,494	**1.92**
Miscellaneous	17,602	10,375	**1.70**
Workload	15,683	9,387	**1.67**
Fairness	4,258	3,567	**1.19**

Table 2. Worked Example, Review #14848.

Section 1: Clause Extraction and Categorization

Clause	Topic	Sim	Top 2 Emotions
dude is huge	*Miscellaneous*	0.030	admiration=0.044, approval=0.024
i mean huge	*Miscellaneous*	-0.010	approval=0.018, annoyance=0.015
is a rude guy	*Miscellaneous*	0.200	anger=0.567, annoyance=0.361
he knows alot about politics	Instr. Effectiveness	0.270	approval=0.293, admiration=0.031

Section 2: Baseline vs. Attenuated Feature Vector

Topic	Baseline Top 4 Emotions	Down-Weighted Top 4 Emotions
Instr. Effectiveness	approval=0.293, admiration=0.031, realization=0.016, optimism=0.004	approval=0.293, admiration=0.031, realization=0.016, optimism=0.004
Fairness	—	—
Workload	—	—
Miscellaneous	anger=0.192, annoyance=0.129, admiration=0.016, approval=0.016	anger=0.063, annoyance=0.042, admiration=0.005, approval=0.005

Section 3: Validity-Weighted Attenuation

D_{misc}	ζ	Raw	Baseline	Attenuated	Δ	Adjusted
0.670	0.330	3.0	2.698	3.125	+0.427	**3.427**

Clause-per-review ratios reveal differences in elaboration depth: *Instructional Effectiveness* averages 1.92 clauses per mentioning review, suggesting multi-clause elaboration, whereas *Fairness* (1.19) tends toward single, concise remarks. *Miscellaneous* and *Workload* fall between these extremes.

Topic co-occurrence patterns show that *Instructional Effectiveness* functions as a central topic: it is the most common standalone topic, and *Workload*, *Fairness*, and *Miscellaneous* each co-occur with it more often than they appear alone.

The rating distribution is negatively skewed, yet low-rating reviews show distinctive patterns: for ratings 1.0–2.0, *Miscellaneous* clauses are the most frequent topic, exceeding the next pedagogical category by approximately 27% in 1-star reviews. *Fairness* clauses also appear disproportionately in low-rating reviews.

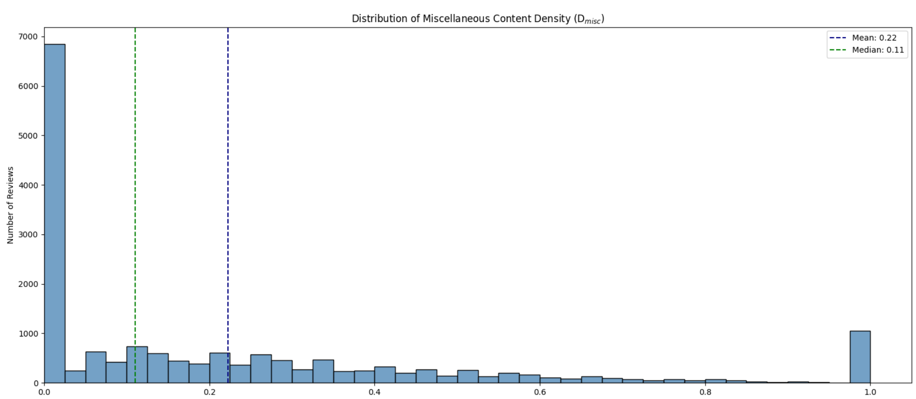

Figure 2. D_{misc} distribution across all reviews.

Figure 2 shows the distribution of D_{misc} across all reviews. The distribution is heavily right-skewed (median 0.11, mean 0.22), with a spike at $D_{\text{misc}} = 1.0$ reflecting reviews composed entirely of non-pedagogical content, a subset receiving maximal attenuation.

Together, these patterns show that pedagogical discourse centres on *Instructional Effectiveness*, that *Miscellaneous* content dominates the lowest-rated reviews, and that the right-skewed D_{misc} distribution motivates proportional rather than uniform attenuation.

4.1.2 Sentiment Extraction

RoBERTa-GoEmotions analysis reveals structured emotional patterns across topics: positive emotions dominate high-rated clauses while negative emotions concentrate in low-rated clauses, though both extremes contain cross-valence signals. *Miscellaneous* emotional probability is consistently weaker than *Instructional Effectiveness* but occasionally exceeds *Fairness* or *Workload*. One-star clauses display a more evenly distributed emotional profile compared to the concentrated signals in 5-star clauses, suggesting that student affect is nuanced and multi-dimensional.

4.1.3 ATC Validation

To assess whether the four-category taxonomy is conceptually distinct, two independent LLMs (GPT-5 mini and Grok 4.1) categorized 386 clauses, yielding Cohen's $\kappa = 0.7273$ ("substantial agreement"; Landis & Koch, 1977), supporting the clarity of category boundaries.

A senior faculty expert with more than 15 years of teaching and evaluation experience independently labelled the same 386 clauses, with multiple valid categories permitted per clause. Under a containment criterion (correct if the model's category appeared in the expert's label set), accuracy is 0.77. Restricting to clauses with exactly one expert label ($n = 287$) yields strict accuracy of 0.72; Table 4 reports per-category precision, recall, and F1.

Table 4. Single-label classification performance of ATC against expert annotations.

Category	Precision	Recall	F1	Support
Instr. Eff.	0.73	0.76	0.74	99
Fairness	0.57	0.76	0.65	17
Workload	0.63	0.78	0.69	54
Miscellaneous	0.82	0.66	0.73	117
Macro Avg	0.68	0.74	0.70	287
Weighted Avg	0.74	0.72	0.72	287
Accuracy		0.72 (n = 287)		

These results indicate acceptable alignment for a proof-of-concept study, though accuracy leaves room for improvement, particularly for *Fairness* and *Workload* where support is limited.

4.1.4 Rating Regression Performance

The dataset is split 80–20 by professor to prevent leakage and better reflect the difficulty of generalizing to unseen instructors. Table 5 reports 5-fold GroupKFold CV and held-out performance; the held-out MAE of 0.6347 falls inside the fold range, confirming consistent generalization.

Table 5. CatBoost 5-fold CV and held-out performance.

	MAE	Spearman ρ	R^2
5-fold CV mean ±SD	0.6300 ±0.0081	0.7284 ±0.0152	0.6255 ±0.0088
Held-out professors	0.6347	0.7352	0.6168

The final model demonstrates low absolute error, strong ordinal consistency, and substantial explained variance on held-out data, suggesting that it captures meaningful relationships between emotional content and student ratings.

To assess directional coherence, SHAP values are computed on held-out professors and grouped into eight polarity × topic categories. High-intensity positive emotions generally increase predicted ratings while high-intensity negative emotions decrease them; however, low-intensity positive emotions can contribute negatively and low-intensity negative emotions positively, indicating that the model responds to the degree of emotional expression rather than its mere presence. These patterns are broadly consistent with expectations, confirming directional coherence while reflecting the inherent variability of real-world sentiment.

4.2 What Attenuation Changes

We evaluate whether attenuation produces structured and interpretable changes rather than arbitrary score adjustments, examining feature importance shifts, correlation changes, and the relationship between attenuation Δs and modulators (D_{misc}, E_{misc}).

4.2.1 Adjustment Outcomes

The statistics in this subsection are descriptive summaries over all 17,127 reviews, not validation metrics. Of these, 10,375 (60.58%) were adjusted, with a mean $|\Delta|$ of 0.1964 and a mean signed Δ of 0.0440. A Wilcoxon signed-rank test confirms the shifts are systematically different from zero ($W = 24{,}260{,}091$, $p < 0.001$).

At the professor level, the overall distribution of average ratings remained largely consistent before and after adjustment.

The three analyses that follow are validation metrics, and are computed on the 206 held-out professors, which were not seen by the CatBoost model. This corresponds to 3,414 reviews for the feature-importance analysis and, of those, the 2,052 containing *Miscellaneous* content for the correlation and modulator analyses, since reviews with $D_{misc} = 0$ are left unchanged by attenuation by construction. These counts are therefore smaller than the full-corpus figures reported above.

4.2.2 Feature Importance Changes

Figure 3 compares baseline and attenuated SHAP-based feature importance. Pedagogically relevant categories increased in importance while *Miscellaneous* sentiment decreased, consistent with the design goal of redirecting model sensitivity away from content classified as construct-irrelevant.

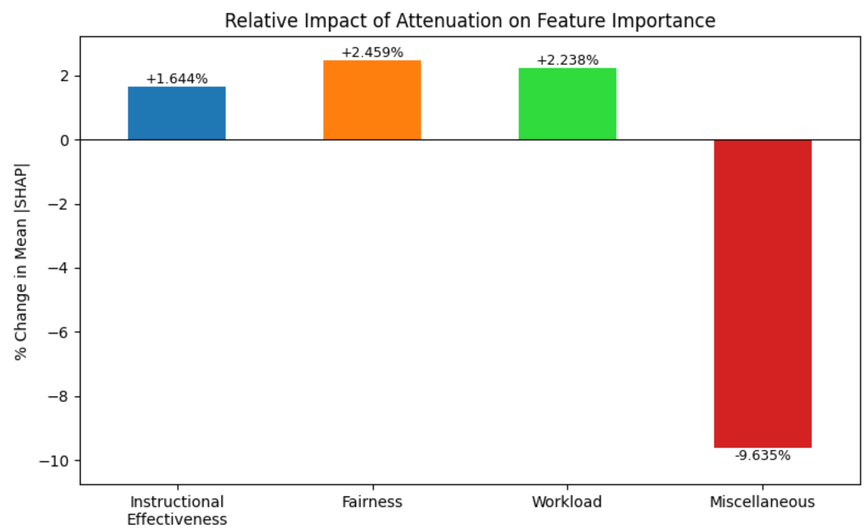

Figure 3. Relative changes in SHAP-based feature importance after attenuation, computed on held-out professors ($n = 3{,}414$ reviews).

4.2.3 Correlation Changes After Attenuation

We compare Pearson and Spearman correlations between baseline emotion features and raw versus adjusted ratings on the 2,052 held-out reviews containing *Miscellaneous* content. The emotion side is held fixed at baseline values in both comparisons; only the rating changes from y_i^{raw} to $\hat{y}_i^{\text{adjusted}} = y_i^{\text{raw}} + \Delta_i$, so any correlation shift is attributable to the adjustment alone.

As shown in Figure 4, attenuation reduces the correlation of *Miscellaneous* features with ratings while maintaining or slightly increasing correlations for pedagogically relevant features, consistent across both linear and rank-based measures.

4.2.4 Modulator Correlations

Table 6 reports correlations between attenuation Δs (separated by sign) and the modulators D_{misc} and E_{misc} on held-out professors.

Table 6. Correlation analysis: attenuation Δ versus modulators, computed on held-out professors ($n = 2{,}052$ reviews).

Comparison Relationship	Pearson r	Spearman ρ
Δ^+ vs. D_{misc}	0.6872	0.6142
Δ^- vs. D_{misc}	-0.7077	-0.6001
Δ^+ vs. Neg. E_{misc}	0.4008	0.4327
Δ^- vs. Pos. E_{misc}	-0.2470	-0.1125

D_{misc} is the primary driver of attenuation magnitude: ratings increase or decrease in proportion to off-topic density, while

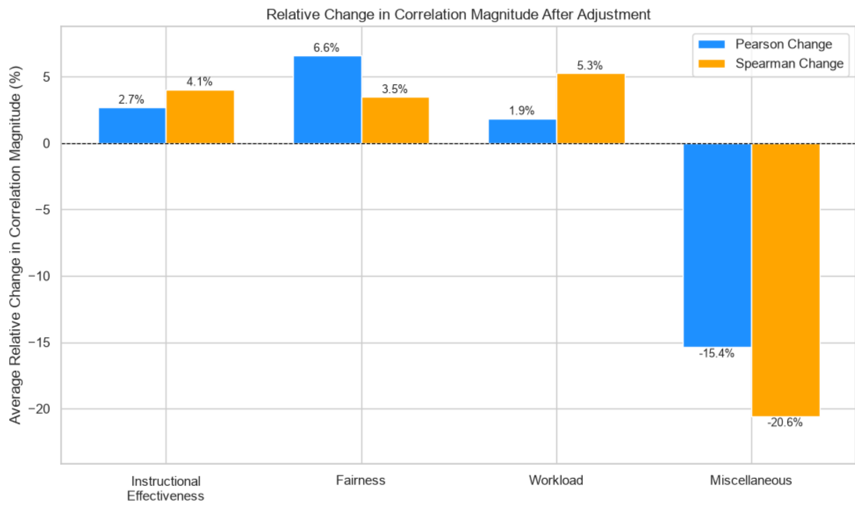

Figure 4. Average relative change in correlation magnitude (Pearson and Spearman) after adjustment, held-out professors ($n = 2{,}052$). Bars show mean $(|\mathrm{corr}_{\mathrm{adjusted}}| - |\mathrm{corr}_{\mathrm{raw}}|)/|\mathrm{corr}_{\mathrm{raw}}|$ across polarity groups.

correlations with E_{misc} intensity are weaker but not insignificant. This aligns with expectations, as D_{misc} directly controls the weighting strength ζ.

A Mann-Whitney U test comparing absolute adjustments between low-noise ($D_{misc} < 0.2$, $n = 690$) and high-noise ($D_{misc} > 0.5$, $n = 508$) reviews confirms significantly larger adjustments for high-noise reviews ($U = 313{,}494$, $p < 0.001$, rank biserial $|r| = 0.789$).

Together, feature importance shifts, correlation changes, and modulator relationships indicate that attenuation introduces structured and interpretable adjustments rather than arbitrary changes.

4.3 Initial Validity and Robustness Evidence

4.3.1 Expert Paired Comparisons

[TODO: Insert the multi-expert consensus results table (overall / coarse / fine accuracy with 95% Wilson intervals; Fleiss' κ) and the interpreting paragraphs. State the 77-of-80 accounting once. The source manuscript's single-expert results are deliberately not transferred here — see the comment block above.]

4.3.2 Permutation Control

Table 7 reports observed values against their null distributions under 1,000 permutations of D_{misc}. The permutation shuffles non-zero D_{misc} values among reviews, preserving the marginal distribution while breaking the correspondence between a review's measured off-topic density and the attenuation it receives. Permuted deltas are then correlated against the *true* densities, testing whether the mechanism's adjustments track real off-topic content or would produce structured-looking results under arbitrary density assignments.

Table 7. Permutation control: observed statistics against the null distribution ($N = 1{,}000$ permutations). p-values are one-sided with the $(k+1)/(N+1)$ correction.

Metric	Observed	Perm. mean	SD	Range	p
Expert accuracy (coarse)	0.8837	0.7891	0.1197	[0.333, 1.000]	0.233
Expert accuracy (fine)	0.7059	0.5772	0.0517	[0.400, 0.739]	0.007
Δ^+ vs. D_{misc} (r)	0.6872	0.2327	0.0291	[0.128, 0.324]	< 0.001
Δ^- vs. D_{misc} (r)	-0.7077	-0.1579	0.0374	$[-0.274, -0.048]$	< 0.001

The modulator correlations provide the cleanest test: no permutation reached or exceeded the observed Δ^+ or Δ^- correlation ($p < 0.001$, both directions). When D_{misc} is decoupled from the review it belongs to, attenuation deltas no longer track true off-topic density.

Expert accuracy separates by condition as the rationale predicts. In the coarse bin, the permuted null frequently reaches the observed value ($p = 0.233$). This is expected: coarse pairs differ so visibly in off-topic content that the CatBoost prediction — which retains all emotion and topic features — often points in the correct direction even when ζ is driven by the wrong density. Coarse accuracy thus functions as a manipulation check confirming that the mechanism responds to gross differences in construct-irrelevant content, but it does not isolate the contribution of D_{misc} specifically. In the fine bin, where paired reviews are closer in off-topic content and the attenuation signal is the tiebreaker, accuracy drops meaningfully under permutation ($p = 0.007$), confirming that discriminative power in close cases depends on the actual review-level density.

The control does not establish that D_{misc} is a valid measure of construct-irrelevant content, nor that the taxonomy underlying it is correct. It establishes a narrower claim: conditional on the ATC classifications, the attenuation mechanism's behaviour is not an artifact of its functional form.

4.3.3 Sensitivity to Attenuation Exponent

The linear attenuation function $\zeta = 1 - D^{s}_{\text{misc}}$ uses $s = 1.0$, operationalizing Huber (1964)'s principle that contaminated observations should be downweighted in proportion to their contamination. The sensitivity analysis below confirms that the framework's reported results do not depend on this choice. We evaluate across $s \in \{0.5, 0.75, 1.0, 1.25, 1.5\}$, symmetric around the operating point.

Table 8 reports pairwise agreement between the attenuation deltas produced at each s. Adjacent values yield near-identical adjustments (e.g., $s = 0.75$ vs. 1.0: $r = 0.995$, mean $|\Delta_a - \Delta_b| = 0.029$), and even the most distant pair ($s = 0.5$ vs. 1.5) retains $r = 0.964$. The maximum single-review difference for that pair is 1.148, but this is one extreme case; the mean across all 2,052 reviews is 0.083.

Table 8. Pairwise agreement of attenuation deltas across values of s (held-out professors, $n = 2{,}052$ reviews with $D_{\text{misc}} > 0$).

s_a	s_b	Pearson r	Spearman ρ	Mean $\lvert\Delta_a - \Delta_b\rvert$	Max $\lvert\Delta_a - \Delta_b\rvert$
0.50	0.75	0.991	0.917	0.043	0.407
0.50	1.00	0.980	0.824	0.063	0.836
0.50	1.25	0.971	0.755	0.074	1.026
0.50	1.50	0.964	0.689	0.083	1.148
0.75	1.00	0.995	0.920	0.029	0.429
0.75	1.25	0.990	0.848	0.043	0.619
0.75	1.50	0.985	0.781	0.052	0.742
1.00	1.25	0.997	0.917	0.022	0.265
1.00	1.50	0.994	0.853	0.031	0.372
1.25	1.50	0.998	0.919	0.017	0.173

Table 9 shows that the modulator correlations — the primary internal validity signal — remain stable across all tested exponents. Pearson r is effectively constant, while Spearman ρ rises monotonically with s as rank ordering concentrates on the measured $D_{\text{misc}} = 1$ tail; in both cases the directional behaviour of Δ^{+} and Δ^{-} is a property of the attenuation concept, not of the particular exponent.

Table 9. Modulator correlations across values of s (held-out professors).

s	Δ^{+} Pearson r	Δ^{+} Spearman ρ	Δ^{-} Pearson r	Δ^{-} Spearman ρ
0.50	0.677	0.505	−0.688	−0.502
0.75	0.684	0.576	−0.703	−0.542
1.00	0.687	0.614	−0.708	−0.600
1.25	0.686	0.656	−0.709	−0.646
1.50	0.677	0.680	−0.717	−0.694

Expert paired-comparison accuracy likewise varies only modestly across s (range: 80.5%–85.7%, $n = 77$ pairs), confirming that expert-judged direction-of-change is not sensitive to the exponent.

4.3.4 Bootstrap Professor Stability

To assess whether the modulator correlations are driven by a small number of outlier professors, we bootstrap-resample the held-out professor set ($n_{\text{boot}} = 1{,}000$ iterations, sampling professors with replacement). Table 10 reports the resulting distributions.

Table 10. Bootstrap 95% confidence intervals for modulator correlations (1,000 resamples of held-out professors).

Metric	Mean	SD	95% CI	
Δ^+ vs. D_{misc} (Pearson r)	0.686	0.018	0.649	0.721
Δ^- vs. D_{misc} (Pearson r)	−0.708	0.018	−0.743	−0.674
Δ^+ vs. D_{misc} (Spearman ρ)	0.614	0.027	0.565	0.662
Δ^- vs. D_{misc} (Spearman ρ)	−0.601	0.025	−0.647	−0.558

All four correlations show narrow confidence intervals (SD ≤ 0.027), indicating that the modulator relationships are distributed broadly across the professor pool rather than driven by a few influential cases.

5. Discussion

5.1 A Validity-Oriented Analytic Primitive

RQ1 results reveal that *Instructional Effectiveness* is the most frequent and influential pedagogical topic, with the highest elaboration depth (1.92 clauses/review), strongest emotional intensities, and highest mean SHAP value. *Workload* contributed more variably, while *Fairness*, though less frequent, showed distinct behaviour in low-rating contexts.

An asymmetry emerges at the rating extremes: 5-star reviews are driven by concentrated admiration and approval toward *Instructional Effectiveness*, whereas 1-star reviews exhibit diffuse negative emotions spread across all topics, including non-pedagogical ones. The disproportionate representation of *Fairness* clauses in low-rating reviews suggests that perceived grading inequity may be a heightened concern among dissatisfied students.

SHAP analysis confirms this hierarchy: Positive *Instructional Effectiveness* was the dominant driver of predicted ratings, followed by Negative *Instructional Effectiveness* and Negative *Miscellaneous*, with *Fairness* and *Workload* contributing least. Emotional responses to pedagogical topics are the primary systematic drivers of rating variation.

Non-pedagogical emotional content also contributes substantially to numerical ratings, aligning with Messick's (1995) concept of CIV. This finding is consistent with broader validity concerns documented in formal SET contexts (Marsh & Roche, 1997; Langbein, 1994; Schiekirka & Raupach, 2015).

Negative *Miscellaneous* ranked third in feature importance (mean SHAP $= 0.674$), exceeding both *Fairness* and *Workload* in either polarity, supporting observations that SET scores frequently capture affective noise rather than instructional quality (Dowell & Neal, 1983; Wongsurawat, 2011; Li et al., 2025).

High ratings exhibit a favourable signal-to-noise ratio because the pedagogical signal is strong, though absolute *Miscellaneous* noise remains high. Low ratings present the inverse: *Miscellaneous* sentiment makes up a proportionally larger share. Construct-irrelevant noise contaminates ratings across the spectrum, motivating auditable attenuation.

Addressing RQ2, internal validation provides tentative evidence that attenuation reduces the influence of CIV while preserving pedagogical signal. Following attenuation, SHAP-based feature importance shifted away from *Miscellaneous* sentiment (-9.6%) and toward pedagogical categories (*Instructional Effectiveness*: $+1.6\%$; *Fairness*: $+2.5\%$; *Workload*: $+2.2\%$).

Correlation analysis confirmed that adjusted ratings maintained or strengthened associations with pedagogical emotions while reducing associations with *Miscellaneous* emotions. Unlike data-exclusion approaches, the method applies proportional down-weighting based on D_{misc}, preserving the raw rating and adding only the model-estimated correction.

The average absolute rating change was 0.1964 points, with larger targeted adjustments for reviews exhibiting high *Miscellaneous* density. The overall professor-level distribution shape remained preserved, consistent with the conservative correction strategy advocated by Marsh and Roche (1997).

RQ3 asks whether the mechanism exhibits behaviour consistent with its validity rationale across independent checks. The modulator correlations provide the strongest evidence: Δ^+ correlates positively with D_{misc} ($r = 0.687$) and Δ^- correlates negatively ($r = -0.708$), confirming that attenuation increases ratings for positively-rated reviews with high off-topic density and decreases ratings for negatively-rated ones. These relationships hold on held-out professors, remain stable under bootstrap resampling (95% CI width ≤ 0.08 for all four correlations), and collapse entirely under permutation of D_{misc} ($p < 0.001$), ruling out the possibility that the functional form alone produces structured-looking results.

Expert paired comparisons provide external validation: on 77 review pairs differing in off-topic density, the expert agreed with the direction of attenuation in 62 cases (80.5%). Fine-grained accuracy, where paired reviews are close in off-topic content and the attenuation signal is the tiebreaker, degrades significantly under permutation ($p = 0.007$), confirming that discriminative power in close cases depends on actual review-level density rather than other features. The sensitivity analysis across $s \in \{0.5, 0.75, 1.0, 1.25, 1.5\}$ further demonstrates that these results are not contingent on the choice of attenuation exponent.

Together, these checks do not establish that the taxonomy or the density measure are correct. They establish a narrower but necessary claim: conditional on the ATC classifications, the mechanism behaves as its validity argument predicts, and that behaviour is reproducible across professors, robust to resampling, and not an artifact of the functional form.

This work extends prior descriptive SET studies toward validity-informed intervention on a learning-analytics trace. By connecting Messick's (1995) unified validity theory with classical reliability principles (Cronbach, 1951; Huber, 1964), the framework translates established psychometric justifications into a concrete computational mechanism that attenuates the modelled contribution of construct-irrelevant affect under a stated pedagogical taxonomy. It addresses similar CIV concerns to those documented by Zhai et al. (2021) in controlled assessments, but in the considerably more challenging setting of informal, unmoderated SET reviews. Each adjustment is traceable to specific topic-level emotions and content density, as illustrated in the worked example (Table 2); given the incomplete specification of pedagogical quality, this inspectability is essential (Doshi-Velez & Kim, 2017). Outputs must remain explainable and contestable (Drachsler & Greller, 2016), connected to pedagogical interpretation and action (Gašević et al., 2015). This positions the framework as initial method evidence for a validity-oriented direction in teaching-feedback analytics, rather than a ready intervention.

5.2 Implications for Responsible Interpretation of Teaching Feedback

If this direction matures beyond proof of concept, it enables specific interpretive acts for multiple stakeholder groups. For educators and teaching-support staff, the framework surfaces raw feedback alongside its topic-level decomposition and adjustment trail. An instructor reviewing their evaluations could distinguish between pedagogically actionable feedback, criticism of explanation clarity, concerns about grading fairness, and broader affective content such as expressions of personal attachment or frustration directed at topics outside the stated pedagogical taxonomy. The adjustment trail makes visible why a rating shifted, allowing the instructor to assess whether the adjustment is meaningful in their context rather than accepting it uncritically. This supports pedagogical reflection informed by analytics rather than replaced by it.

At an aggregate level, topic-specific emotional profiles could be examined across instructors, courses, or departments. Concentrated negative sentiment directed at Fairness across a department's evaluations might surface a systemic concern about grading practices, whereas the same pattern in a single instructor's reviews might inform targeted pedagogical support. These patterns are already present in the framework's outputs and require no additional computation, only decisions about the appropriate level of aggregation under which such information is shared. For students navigating informal platforms like RMP, the framework offers additional context for interpreting ratings. Rather than relying on a single numerical score, a student could see how much of a review's content is pedagogically focused, which topics drive the emotional signal, and how the rating relates to the pedagogical substance of the feedback. A professor whose low rating is driven primarily by Workload sentiment presents a different picture from one whose rating reflects Instructional Effectiveness concerns, a distinction the raw number does not make.

In all cases, raw ratings and reviews are preserved in full. The framework adds a layer of interpretation; it does not replace or conceal anything. However, additional information is not inherently safer to act on; the constraints on interpretation discussed in the following section apply to all of these use cases.

5.3 Boundaries, Risks, and Non-Use

The framework produces model-based adjustments under a stated pedagogical taxonomy. An attenuated rating is not a validated estimate of teaching quality, and treating it as one misrepresents both the method and its evidence base. Equally, the framework decomposes a review into topic and emotion components, it does not summarize or replace the review itself. Any interpretation that bypasses the original text discards exactly the context the audit trail is designed to preserve.

Auditability alone does not make a deployment trustworthy. Drachsler and Greller's DELICATE framework treats transparency as one element of trusted learning analytics, alongside an explicit purpose, stakeholder involvement, data governance, and routes to contest harmful consequences. Any platform or institution presenting these outputs would be responsible for those conditions; this proof of concept provides no such deployment model (Drachsler & Greller, 2016).

A distinction must be drawn between automated calculation and automated deployment. Computing attenuated ratings and decompositions programmatically is the intended use, the mechanism is deterministic and reproducible. Deploying those outputs into a decision pipeline that acts without human interpretation is not. The framework is designed to inform a reader, not to replace one.

The framework addresses one source of construct-irrelevant variance: the modelled contribution of affect in content classified outside the stated taxonomy. Other documented sources of bias in teaching evaluations — including gender, race, and popularity effects, remain present in the data and in the adjusted ratings. A user who assumes that attenuation has produced a "cleaned" rating risks greater confidence in a result that still carries unaddressed contamination.

Responsible use requires understanding where the framework's auditability ends. The attenuation rule itself is fully transparent: $\zeta = 1 - D_{\text{misc}}$ is a fixed, monotonic function with no learned parameters. However, the components that produce its inputs are not equally interpretable. Topic assignment relies on sentence embeddings and cosine similarity thresholds, emotion classification uses a fine-tuned RoBERTa model, and the downstream rating model is a CatBoost ensemble. The audit trail shows *what* the framework did — which clauses were classified where, what emotions were detected, how the rating changed — but a user contesting an adjustment must recognize that the upstream classifications themselves are model outputs, not ground-truth labels. Engaging critically with the trail, rather than accepting it as a complete explanation, is part of using the framework responsibly.

6. Limitations and Future Work

While the framework demonstrates promising proof-of-concept results, key limitations should be acknowledged.

The absence of an objective ground truth for teaching quality against which adjusted ratings can be validated is a fundamental limitation of this research. While the framework demonstrates internal consistency, improved signal-to-noise ratios, and expert alignment, these are indirect forms of validation. They do not definitively prove that adjusted ratings better reflect true teaching quality. Establishing stronger criterion validity would require external benchmarks such as student learning outcomes, peer evaluations, or longitudinal performance data, which are not available for informal SET platforms such as RMP. As a result, the framework should be interpreted as a method for reducing noise in SET ratings rather than a definitive correction toward true teaching quality. This limitation reflects a broader challenge in evaluating teaching performance, where the underlying construct cannot be directly observed. In addition to this fundamental limitation, several data-related and methodological constraints warrant discussion.

The exclusive use of RMP data limits the generalizability of these findings. RMP reviews are anonymous, unmoderated, and self-selected, often differing substantially from formal SET instruments in structure, administration, and respondent motivation. Furthermore, the RMP dataset is negatively skewed, containing many more 5-star ratings than other ratings, which may bias model predictions and limit generalization. Data cleaning and clause segmentation are also not fully reliable; irregular punctuation and multiple clauses in a single sentence sometimes cause inaccurate splitting, affecting downstream analysis. Clause-level sentiment aggregation may lose nuance, particularly in short reviews with mixed sentiment.

The Aspect-Term Categorization (ATC) relies on informally defined, broad topics and assigns each clause to only one topic, even when multiple topics might apply. This reduces accuracy for multi-topic clauses and can result in the misclassification of topics. While ATC accuracy was acceptable for a zero-shot approach, it leaves room for improvement. Additionally, the RoBERTa-GoEmotions model was trained on Reddit comments rather than student reviews, which can lead to the misinterpretation of sarcasm or informal language, reducing accuracy for context-specific sentimental expression.

The attenuation mechanism itself has specific constraints. When *Miscellaneous* content is the only topic present in a review, $D_{misc} = 1$. This results in fully down-weighted emotions, producing identical feature vectors for different reviews. In these cases, the model repeatedly predicts the same attenuated rating. Furthermore, attenuation outcomes heavily depend on the regression model. Some reviews with high D_{misc} receive minimal adjustment, whereas others with lower D_{misc} receive relatively larger adjustments. Results may also be sensitive to the empirically selected thresholds and hyperparameters used in ATC and regression. In addition, the framework cannot effectively attenuate ratings for reviews with neutral emotions but high *Miscellaneous* content.

A critical scope limitation is that the framework addresses only one specific type of CIV: non-pedagogical emotional content or affective noise. It does not mitigate other known sources of bias in SET, such as grading leniency, course difficulty, or demographic biases. The dataset contained demographic information which was not utilized in this research, meaning adjusted scores may still reflect gender or racial inequities present in the raw data.

Expert attenuation validation is similarly limited, measuring only the agreement in relative adjustments without capturing the accuracy of specific adjustment magnitudes. While large attenuation Δs strongly align with expert judgment, smaller adjustment differences are less reliably captured, reducing confidence in subtle cases.

[TODO: The final limitations paragraph in the source manuscript concerns reliance on a single expert annotator. It is not transferred here because the multi-expert consensus supersedes it. Replace with a statement of the remaining limits of the expanded expert sample once those results exist.]

Future work should evaluate external validity using larger datasets derived from formal SET instruments. Institutional evaluations often employ structured prompts and standardized Likert scales, sometimes linked to student ID, which might naturally reduce the proportion of *Miscellaneous* content, potentially requiring the recalibration of attenuation parameters for a different noise baseline.

Regardless of the modelling approach, the current ATC taxonomy remains relatively broad. A more granular category structure defined by domain experts, or aligned with SEEQ definitions (Marsh, 1982) could capture important pedagogical distinctions. For example, distinguishing between *procedural fairness* (grading policies and rubrics) and *interactional fairness* (empathy and respect). Moving from single-label classification toward multi-label or soft-assignment approaches would also better reflect how individual clauses frequently address multiple topics simultaneously.

Future work should co-design a formal-SET version of the framework with educators, students, and institutional stewards, grounding custom taxonomies in course learning design and specifying the purpose, access, retention, and contestation arrangements required before deployment.

The current density metric (D_{misc}) relies primarily on word counts, which serve only as a proxy for the informational relevance of a review. Alternative approaches could incorporate semantic similarity with expert-defined pedagogical descriptors or other more sophisticated measures of textual quality to provide better estimates of construct relevance.

Additionally, while the attenuation mechanism is here evaluated on professors held out from model fitting, this remains an internal split of a single corpus. Future work should evaluate the mechanism on an independent dataset drawn from a different institutional or platform context, and against a formal external criterion of teaching quality, to establish how far the reported behaviour generalizes beyond the population it was developed on. Establishing the framework as evidence-based under evaluative standards such as those proposed by Ferguson and Clow (2017) requires external criterion validation and replication across institutional contexts.

Equally important are fairness audits that ensure the framework does not unintentionally amplify existing inequities. These audits should examine system behaviour across instructor demographics and disciplinary contexts.

7. Conclusion

Open-ended teaching feedback is an increasingly common learning-analytics trace, yet computational approaches to this data rarely move beyond description toward validity-informed intervention. This research presented validity-weighted attenuation, a framework that mitigates construct-irrelevant variance in teaching evaluations by producing adjusted ratings that attenuate the influence of content classified outside a stated pedagogical taxonomy, in order to complement and extend existing analytics available to students, educators, and institutional decision-makers.

The framework moves beyond descriptive sentiment analysis toward active, interpretable mitigation of construct-irrelevant variance. Its three primary contributions reflect this: a novel attenuation mechanism that directly intervenes on ratings rather than merely detecting noise, where each adjustment is traceable to specific topic-level emotions and content density; a theoretically grounded design that translates established psychometric principles from Messick (1995), Cronbach (1951), and Huber (1964) into a concrete computational mechanism; and initial empirical proof-of-concept validation on a large, held-out-instructor sample with expert and robustness evidence.

Crucially, the framework preserves every original review and its full emotional profile; attenuation reweights how affect contributes to a summary rating, it does not silence student voice. This design choice reflects a commitment to contestability: stakeholders can inspect, question, and override any adjustment. As a proof of concept, the work demonstrates that validity concerns in large-scale teaching feedback can be computationally addressed while balancing interpretability with performance, establishing a foundation for validity-aware analytic tools that decouple pedagogical signals from broader student experience under a taxonomy that stakeholders themselves define, so the interpretive lens reflects their priorities rather than a fixed computational judgment.

Declaration of Conflicting Interest

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

[TODO: Funding statement, or a statement that no funding was received.]

Ethical Considerations

All expert labelling was conducted under a REB exemption approved by the authors' institution, ensuring that annotation procedures adhered to ethical research standards.

Data & Code Availability

A public codebase for this framework is available online. Due to privacy restrictions, the original RMP dataset is not included; however, all fully anonymized expert-labelled data used for validation is provided.

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